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ENROLLEASE: DEVELOPMENT OF A FAMILY VILLAGE RESOURCES ELEMENTARY SCHOOL WEBSITE WITH AN INTEGRATED ENROLLMENT SYSTEM

A Thesis

presented to the

Faculty of Eulogio "Amang" Rodriguez

Institute of Science and Technology Cavite Campus

In Partial Fulfillment

of the Requirements for the Degree

Bachelor of Science in Computer Science

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2025

CHAPTER 1

The Problem and Its Background

Introduction

Technology became more advanced and systematized with computers and electronics. It continuously evolves day-by-day and adapting in today is constantly changing the digital world. Students who use the Internet report that it is the primary method for completing school work (Leu et al., 2003). Students use the Internet to research information, conduct surveys, collect data, create presentations, and communicate with their teachers and classmates (Cuban, 2001; Wenglinsky, 2005). Students who are comfortable with the use of technology are more engaged in classroom activities and are generally more efficient.(Bosco, 2001; Kulik, 2003;). Elaborating more on the functions and features of the project, the School website for F.V.R Elementary School is a stand-alone website comprising three modules: student, teacher, and parents, which are all required to log-in to the system for them to be able to use it.

Theoretical Framework

The development of the school website for an

elementary school in the Philippines is based on key educational and technological theories. The Constructivist Learning Theory supports the use of interactive online tools to promote active student engagement. The TPACK Framework ensures that the technology used aligns well with effective teaching strategies. Diffusion of Innovation Theory guides the smooth adoption of the website by stakeholders, while Systems Theory emphasizes the website's role in improving communication and coordination within the school. Media Richness Theory highlights the need for engaging multimedia content suitable for young learners, and Culturally Responsive Pedagogy ensures the website reflects Filipino culture and values. Together, these theories support.

Conceptual Framework

This study is guided by the model Input, Process and Output presented in the 1st figure.

Frame 1 involves gathering all necessary information and resources to initiate the system's functionality. This includes student information such as personal and academic details needed for enrollment, as well as teacher and administrator accounts to manage system operations.

Enrollment data is submitted through online forms, accompanied by school announcements and multimedia content like images or videos to enrich the user interface. Technical inputs like Firebase configuration are essential for backend operations, ensuring secure authentication and real-time data processing. Additionally, user login credentials are required to control access, and curriculum content provided by users supports academic functions within the platform.

Frame 2 performs a series of automated operations that convert inputs into functional outcomes. User authentication is handled via Firebase, validating login credentials and granting secure access to the platform. Once authenticated, the system retrieves data from the database and displays it accordingly, ensuring that each user sees only the information relevant to them.The submitted forms go through validation to confirm that all fields are correctly filled and documents are complete. Based on user roles, personalized dashboards are generated, and authorized users can manage content such as announcements and enrollment details. The system supports real-time updates, so any changes made are instantly

reflected across the platform, maintaining accuracy and efficiency.

Frame 3 delivers the final results of the processed data. Users are provided with personalized dashboards tailored to their roles, such as student, teacher, or administrator. Successful applications result in enrollment confirmations, while school announcements are displayed promptly to keep users informed. The platform showcases updated school information and academic content, supported by an appealing and organized visual presentation. Pages are designed to be responsive, offering seamless access across various devices. Login feedback also guides users by confirming successful access or alerting them to errors, contributing to a smoother user experience.

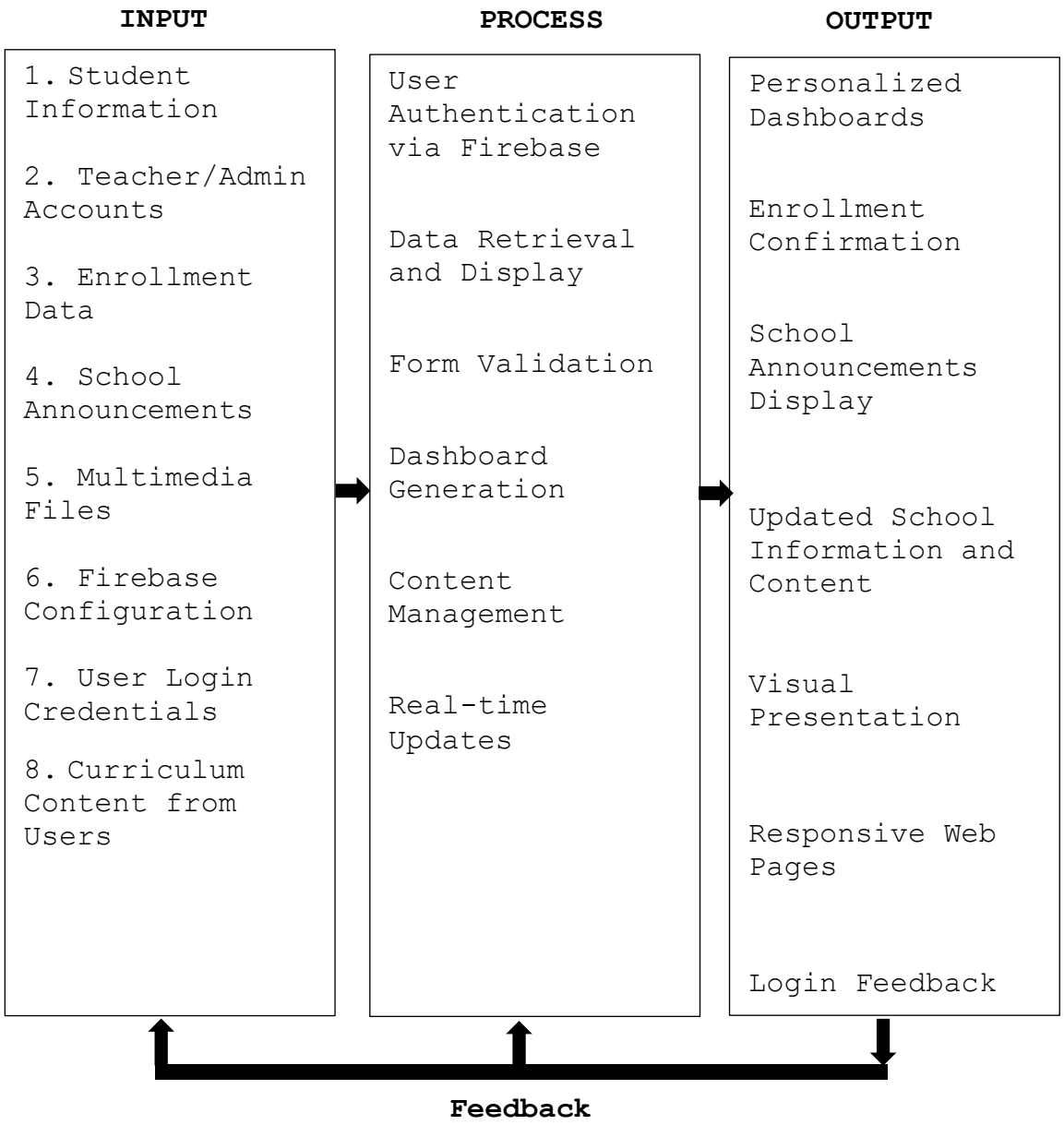

Figure 1: Conceptual Framework of the Study

Statement of the Problem

The study aims to develop a website for Family Village Resources Elementary School with an integrated enrollment system to improve the enrollment process and online accessibility.

1. What are the common challenges encountered in the current enrollment process of Family Village Resources Elementary School?
2. What are the requirements needed in developing the website with an integrated enrollment system in terms of:

   2.1. USER INTERFACE DESIGN;

   2.2. FUNCTIONALITY;

   2.3. DATA SECURITY; AND

   2.4. USER ACCESSIBILITY

3. How effective is the developed system in addressing the identified challenges in the enrollment process?
4. Based on the findings, what recommendations can be proposed to further improve the system?

Hypothesis

The developed website with an integrated enrollment system will improve the efficiency, accessibility, and overall user experience of the enrollment process in Family Village Resources Elementary School.

Scope and Limitation of the Study

The scope of the study includes the design and development of a website with an integrated enrollment system for Family Village Resources Elementary School. The study focuses on enhancing user interface design, system functionality, data security, and accessibility for both parents and school administrators. The study is limited to the enrollment process only and does not include other school management systems such as grading, attendance tracking, and financial transactions. Other features outside of the enrollment process will not be covered by this study.

Significance of the Study

This study aims to develop and implement a functional website with an enrollment system for Family Village

Resources Elementary School (FVR ES) located in GMA, Cavite for a easier school administration, enhancement in enrollment procedures, communication with parents & the community including other scholastic metrics. This study will be useful to the following stakeholders:

Students They will gain access to a centralized online portal, streamlining their access to school-related data and resources.

Parents and Guardians They will benefit from a more efficient and convenient online enrollment process, as well as simplified access to school activities, updates on student progress, important school notices, and direct communication channels with teachers.

Teachers and Staff This system promises to streamline enrollment, thereby reducing manual labor and improving the methods for managing data.

School Administration They will acquire a modern instrument for managing enrollment, effectively communicating, and enhancing the school's operational efficiency and public image.

Future Researchers The study will serve as a valuable case study and provide insights into the development and deployment of integrated school websites and enrollment systems within a Philippine elementary school setting.

Definition of Terms

For better understanding and interpretation of this study, the following terms are operationally defined:

Enrollment System A systematic procedure, often facilitated by a software application, utilized by educational institutions to manage the processes of student admission and registration.

Integrated System A system where various components or modules function in unison, sharing data and functionality. In this context, it refers to the combination of the website and the enrollment system into a single platform.

Online Platform A web-based environment or system accessible via the internet, offering a spectrum of services and information.

School Administration The individuals or department accountable for overseeing the operational aspects of the

school.

Streamlined Process A process that has been made more efficient and less complex, resulting in fewer unnecessary steps and reduced workload.

Stakeholders Individuals or groups with an interest in or affected by the project or study outcomes. This includes students, parents and guardians, teachers and staff, school administration, and future researchers.

CHAPTER 2

Review of Related Literature and Studies

This chapter contains the information derived from the different sources such as books, electronic books and internet sources to emphasize and give support to the ideas and views regarding the research. This also includes the related literature and studies both foreign and local which are relevant to the study.

Local Literature

Estanislao et al. (2020) piloted an E-Learning Outreach Program for public elementary schools, providing insights into the basic infrastructure and digital literacy required for successful e-learning platform deployment. The findings highlighted how the provision of internet-connected e-learning stations significantly improved students' digital competence and readiness for web-based education systems. This suggests that setting up a school website could also serve as a foundation for broader e-learning initiatives.

De Guzman and Cruz (2019) emphasized the vital role of websites in improving communication between the school

and parents. The findings showed that elementary schools which adopted digital platforms like websites experienced more engaged and informed parents. The authors noted that websites allowed parents to monitor school activities, events, and even student progress despite their busy schedules. Furthermore, the study suggested that digital communication platforms foster a more collaborative relationship between schools and families.

Mateo (2021) investigated the results and revealed that digital literacy among teachers plays a crucial role in the maintenance and effectiveness of school websites. Mateo found that when teachers were trained in basic website management, they became more proactive in updating content and ensuring the website's relevance. This study implies that capacity-building programs focused on ICT skills are vital for the successful implementation and sustainability of school websites.

Villanueva (2017) discussed how the integration of ICT programs helped in launching school websites. She emphasized the supportive role of DepEd's ICT initiatives, which encouraged schools to digitize communication and administrative services. Villanueva observed that schools

with active websites also enhanced their teaching strategies by providing online learning resources and announcements, thus promoting a more modern educational environment even at the elementary level.

Garcia (2020) said that apart from reading, writing, and arithmetic, media literacy has become one of the necessary literacy that the children should acquire for them to successfully navigate their ever-changing digital media environments. However, an equally heavy work is needed to advance media literacy advocacy among our younger media consumers-the Philippine basic education curriculum has yet to fully integrate media literacy for elementary and junior high school learners, and we have yet to see more culturally sensitive, child-friendly, and age-appropriate educational materials that focus on enhancing one's media literacy skills and competencies.

Foreign Literature

According to Zen et al. (2023), online registration of new students is a solution for more transparent, easy to collect data, and informative new student registration. Not many schools still implement or use information

systems in the new student registration process.

Onyenwe et al. (2022) said that parents play vital roles in contributing to students' achievement. They need to be actively involved in monitoring their children's status and daily performance in school using web-based system.

Research conducted by **Asrin et al. (2023)** about implementing a web-based school information system at SDN 14 Pontianak City, it reveals that school information systems address challenges related to ineffective internal information delivery within the school.

Limbong et al. (2021) pointed out that information systems can be an effective assistant in the enrollment process carried out by teachers or teaching staff as it lessens the manual works in the school.

Sara et al. (2022) stated that students' grades and information will be easier to manage with the help of a School website.

Local Studies

Based from the study conducted by **Mina et al. (2021),** the students marked the online enrollment system very good

because it is better than the physical enrollment process. The online registration system is far better than the walk-in queues owing to the fact that new technology leads to a change in how people access different services.

According to **Grepon et al. (2021)**, as the number of students grow bigger, it is in need of a systematic enrollment process where face to face and online process should be taken into consideration. The enrollment process could be better and become readily accessible to the users. The procedure will then be conducted at optimum operation, catering more individuals at the same time with the new standard process of enrollment. The most common problems encountered by the school at the end of each semester are the student enrollment process, which typically lasts 2-3 weeks because most of the admission and storage of student information is done manually. Retrieval and generation of data and information is also a concern, as it takes a longer period to consolidate information that is required to complete the course.

Hayagan (2022) stated that Central Philippines State University San Carlos Campus Online Enrollment System was efficient, effective, user-friendly and easy to

understand. The enrollment procedure is a huge help to the student, parent and to the school. The students of CPSU San Carlos Campus easily understands the content and meaning of each enrollment procedures implemented online. The students also agreed that the online enrollment system saves time effectively. The school provides wide access and stores accurate student accounts with the help of the enrollment methods implemented online. Through the help of an online enrollment system, the school can easily track and monitor the needs of the school and the student. The implementation is highly suggested by the respondents to go on due to the efficiency of the said online system.

Caratiquit (2021) stated that a school system or website might be used by academic institutions such as colleges and universities to keep student and staff records easily. Additional benefits include the ability to save money, time, and effort by using a web-based information and publication system for schools. If an institution has a well-designed website, it will be possible to create online portal brochures. The school's management will not have to worry about running print or television advertisements to promote the institution.

Additionally, the necessity that parents and other members of the school community receive printed documents and announcements is gone.

Based from the study conducted by **Bracero et al. (2025)**, the State Universities and Colleges continuously working on the implementation of information systems particularly the enrolment system. This effort among educational institutions to automate enrolment system integrates various processes into an information system which consolidates data into a database server, thus, access to data and information becomes more efficient. The implementation of such systems also lessens time requirement for a certain transaction to finish and therefore increases personnel efficiency and productivity.

Enrollment system is highly usable, secured, efficient, and provides a fast and easy way to manage enrollee records, updating students/enrollee data, viewing the total number of enrolled students, adding of courses, subjects, and class schedules.

Foreign Studies

According to **Nwachukwu and Okafor (2019)**, an online

student admission system for secondary schools in Nigeria allow users to register, submit necessary documents, and receive real-time confirmation through a web-based platform developed using PHP and MySQL. Their findings showed that the system significantly reduced manual labor, human error, and processing time in the enrollment procedure. The relevance of this study to the current research lies in the successful implementation of a digital enrollment process that enhances efficiency and accessibility elements critical in the development of the FVR Elementary School website with integrated enrollment functions.

Badusha and Pavithra (2020) said that an integrated web-based school management system in India. address fragmented school operations, particularly in enrollment and student information management. This system employed Firebase for its real-time capabilities and ensured that users such as parents, teachers, and administrators could interact seamlessly within the platform. Their study emphasized the importance of centralized systems in promoting better coordination and improved data accessibility, supporting the conceptual framework of the

present research.

According to **Phan and Nguyen (2018)** who conducted a study in Vietnam focusing on a web-based application for online enrollment and grade monitoring in elementary schools with the use of HTML, CSS, JavaScript, and Firebase to create an interactive and responsive interface tailored to the needs of students and parents. The outcome demonstrated increased parental involvement, reduced administrative workload, and enhanced real-time data handling principles that align with the goals and technological setup of the current study.

A cloud-based student enrollment system for primary schools in Taiwan that utilized Node.js, MongoDB, and cloud hosting through AWS **by Chen and Li (2021)** revealed that a marked improvement in enrollment accuracy and efficiency while minimizing paper usage. This approach reflects modern cloud practices relevant to the current research, especially in the areas of scalability and data integrity.

Garcia and Torres (2022) who implemented a multilingual school admission platform in Spain to accommodate the country's diverse linguistic demographics

that was built with Django and PostgreSQL, the system allowed applicants to complete the process in their native language, improving user experience and accessibility. Their research showed that localized interfaces boosted application completion rates and user satisfaction principles that align with the user-centered design aspect of the FVR Elementary School website project.

Synthesis

Modern technology significantly alleviates the challenges associated with manual tasks and processes across various fields, including the educational sector. In school settings, the integration of technology streamlines essential services such as enrollment and information dissemination. When developing a functional, efficient, and well-structured system or website, it is crucial to prioritize the needs and experience of the users.

This review of related literature and studies demonstrates that a school website, integrated with an enrollment system offers significant benefits to students, parents, teachers, and school administrators by

effectively facilitating the dissemination of information and streamlining enrollment processes through reduced manual intervention.

CHAPTER 3

Methodology

This chapter presents the research design, the respondents and the sampling technique, the research instruments used in gathering data, the data gathering procedures and the statistical treatments of data.

Research Design

This study employed a descriptive research design using a quantitative approach to evaluate the effectiveness and user satisfaction of the developed web-based enrollment system for Family Village Resources Elementary School (FVR ES). A survey questionnaire was administered to gather structured feedback from parents, teachers, and school administrators. This method allowed the researchers to systematically describe users' experiences, perceptions, and suggestions regarding the system.

Population and Sample

Table 1

Sample Coverage of the Respondents in the Study

Key Groups	No. of Respondents
School Administrators/Registrars	3
Parents or Guardians	20
Teachers	27
Total	**50**

The respondents of the study were selected from three key groups directly involved in or affected by the enrollment process: school administrators/registrars, parents or guardians, and teachers. A total of thirty (50) respondents selected in the study, consisting of three (3) school administrators/registrars, ten (20) parents or guardians, and seventeen (27) teachers.

To ensure that feedback was gathered from individuals with relevant experience and direct interaction with the enrollment system, the researchers employed purposive sampling. This non-probability sampling technique allowed the selection of respondents based on their roles and familiarity with the current enrollment procedures, making

them suitable sources of input for the evaluation and improvement of the proposed system.

Respondents of the Study

The study involved fifty (50) respondents from the three (3) key groups who are involved in the school's enrollment process.

Research Instrument

The primary instrument used to gather the necessary data was a survey questionnaire which was being formulated.

This questionnaire focuses on users satisfaction, experiences and suggestions for improvements. It aims to collect measurable feedback from the parents, teachers, and school administrators who interacted with the website and the online enrollment system.

For the Respondents:

Weighted Mean Range	Verbal Interpretation
5	Strongly Agree
4	Agree
3	Uncertain
2	Disagree
1	Strongly Disagree

Data Gathering Procedure

The following procedure was used in this study. (1) books,(2) online resources,(3) thesis and, (4) other materials were used as primary sources of data to formulate the questionnaire to be given to respondents.The survey focused on usability, accessibility, design and interface and content of the website. The gathered data was organized and prepared for analysis to assess the overall effectiveness of the website and identify the areas of improvement. Moreover, consent from the participants was formally obtained prior to the administration of the surveys at the selected elementary school.

Statistical Treatment of Data

The following statistical methods were applied to analyze the data collected from respondents:

1. Frequency and Percentage was utilized to determine the demographic distribution and responses of the participants regarding the system's usability, accessibility, design, and content.

   Formula:

   $$\text{Percentage } (\%) = \left(\frac{f}{n}\right) x\, 100$$

   f = frequency of each response

   n = total number of respondents

2. Weighted mean was used to assess the respondents' level of satisfaction with the developed website system.

   Formula:

   $$\frac{\sum(f * x)}{n}$$

   f = frequency of each scale point

   x = corresponding scale weight (1 to 5)

   n = total number of responses

Scale	Weighted Mean Range	Verbal Interpretations
5	5	Strongly Agree
4	4	Agree
3	3	Uncertain
2	2	Disagree
1	1	Strongly Disagree